	the same for mud near the centre or edge.	
4(a)	acceleration is directly proportional to its displacement from a fixed point	B1
	acceleration is always in opposite direction to its displacement/always towards the fixed point)	B1
	Marker's comments: Majority of candidates got it wrong.	
4(b)	<table><tr><th></th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th></tr><tr><td>displacement</td><td>+</td><td>0</td><td>–</td><td>0</td><td>+</td></tr><tr><td>velocity</td><td>0</td><td>–</td><td>0</td><td>+</td><td>0</td></tr><tr><td>acceleration</td><td>–</td><td>0</td><td>+</td><td>0</td><td>–</td></tr></table>	
	displacement line correct	B1
	acceleration line opposite to displacement line	B1
	velocity line has B D & F as zero with answers to C and E consistent with their displacement line	B1
	Marker's comments: Candidates demonstrated understanding of vector concepts.	
4(c) (i)	phase difference between displacement and velocity is $\pi/2$ OR $3\pi/2$ radians OR 90°	B1
4(c) (ii)	displacement and acceleration are exactly out of phase OR out of phase by π radians OR 180°	B1
4(d) (i)1	amplitude = 8.0 cm	A1

4(d)(i)2	T = 2.0 s so f = 0.50 Hz	A1
4(d)(ii)1	$F = mA\omega^2$ $= 0.020 \times 0.080 \times (2\pi/2.0)^2 = 0.0158$ N (or 0.016 N)	A1
4(d)(ii)2	negative cosine graph of any amplitude for at least 3.5 s	B1
	Marker's comments: Poorly and wrongly drawn graph. Some negative sine graph were also drawn.	
5(a)	electric field strength is the force acting per unit positive charge	B1